

RECYM: An Automated Multi-Feeder Workflow for Demand Allocation and Load-Flow Studies in CYMDIST Distribution Networks

Mac Tapia Ccosyo

Electro Dunas S.A.A., Peru

e-mail: mtapia@electrodunas.com

Abstract—Distribution utilities need repeatable procedures to clean network models, allocate measured headroom demand, inject important-customer and prospective spot loads, and compare situational versus projected operating cases. This paper presents RECYM, a multi-feeder software suite that couples CYMDIST/CymPy with a seven-step SPA and a batch pipeline. The workflow integrates network diagnostics, large-customer energy/power mapping to SED nodes, kWh-based load allocation via the CYMDIST COM API, locked three-phase spot loads, dual load-flow scenarios, and campaign artifacts. RECYM is wired to the Electro Dunas Access database with 97 feeder configurations and multiple .zxst studies; PA217 (22.9 kV) is the representative experimental sample. On PA217 the campaign achieved zero diagnostic problems, verified customer precision, allocation balance error $\text{eps}_P \sim 7.6\text{e-}5\%$, a 1420 kW spot load, and dual load flows with $\text{delta}_P = +17.17\%$.

Index Terms—CYMDIST, demand studies, distribution networks, load allocation, load flow analysis, power system modeling, software tools.

I. INTRODUCTION

Distribution planning relies on detailed three-phase feeder models [1], [2]. CYMDIST provides allocation and unbalanced load flow [3], [4], yet campaigns often remain manual. OpenDSS and pandapower automate research [11], [12]; theses often migrate CYMDIST models [13]-[15]. RECYM instead orchestrates COM/CymPy on native .zxst/.mdb assets, with a multi-feeder architecture (97 configs), a §§1-7 campaign, and validation on sample feeder PA217.

Contributions: (i) layered architecture L1-L5; (ii) mathematical models for kWh allocation and dual LF with pseudocode; (iii) experimental statistical validation on PA217 as a sample from the Electro Dunas catalog.

II. THEORETICAL FRAMEWORK AND RELATED WORK

A. Three-Phase Flow and Allocation

Kersting [1] and Cheng-Shirmohammadi [9], [10] underpin radial/weakly meshed flow. Energy-based allocation often outperforms connected-kVA when measurements exist [7], [8]. CYMDIST implements Connected+Total templates [4].

B. Positioning

Unlike CYMDIST→OpenDSS conversions [13]-[15], RECYM keeps the system of record. Versus utility integrations [16], [17], it adds SPA/job orchestration, Locked SpotLoad, and situational/projected scenarios [23]-[25].

III. SYSTEM ARCHITECTURE

RECYM is a five-layer stack L1..L5 (Fig. 1): React SPA; FastAPI/SSE API + Flask bridge; domain pipeline/analysis/optimization; CymPy/COM core; .zxst/.mdb/Excel data. Configuration (S, F) with |F|=97 feeders plus ELD study.

Fig. 1. RECYM layered architecture

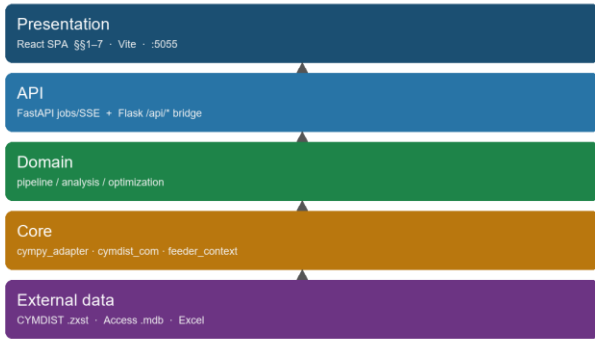


Fig. 1. RECYM layered architecture.

Fig. 2. RECYM campaign workflow §§1-7

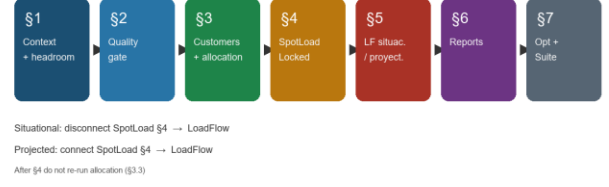


Fig. 2. Operational campaign §§1-7.

IV. MATHEMATICAL MODELS AND ALGORITHMS

A. kWh Allocation Model

Let P_{cab} be headroom active power and $L = F \cup U$ the Locked/unlocked partition. Residual and allocation are $P_{cab_res} = P_{cab} - \sum_{j \in F} P_{j_fix}$ and $P_i = P_{cab_res} \cdot E_i / \sum_{u \in U} E_u$ for $i \in U$. Balance error is $eps_P = |\sum_i P_i - P_{cab}| / P_{cab} \cdot 100\%$. Precision uses MAE_P and $RMSE_P$ of $|\Delta P_i|$ with tolerance 0.01 kW (0.5 kWh for energy). This paragraph states the first closed-form mathematical model [7], [8], [4].

B. Spot-Load and Dual LF Model

For SpotLoad ($P3\phi$, $Q3\phi$): $P\phi = P3\phi/3$, $Q\phi = Q3\phi/3$, $\phi \in \{A, B, C\}$, Locked. With flag $c \in \{0, 1\}$, $y_{sit} = F(x; c=0)$ and $y_{prj} = F(x; c=1)$; $\Delta y = y_{prj} - y_{sit}$; $\delta P = 100 \cdot \Delta P / P_{sit}$. This paragraph states the second mathematical model [9], [10], [23]-[25].

C. Pseudocode

Algorithm 1 (allocation): compute P_{cab_res} ; distribute P_i by E_i ; COM write; return eps_P . Algorithm 2 (dual LF): disconnect SpotLoad → situational LF; connect → projected LF; Δy . Algorithm 3 (campaign): headroom → diagnose $N_{msg}=0$ → Alg.1 → SpotLoad → Alg.2 → artifacts/closure.

V. EXPERIMENTATION

Corpus: 97 configs + ELD, shared Access DB, multiple .zxst. Sample: PA217 (22.9 kV), 1324 nodes, 253 loads. Headroom L-PA235: $P=9537.88$ kW, $Q=2587.65$ kvar. Protocol `dry_run=false` §§1-7. The utility delivery report (doc/informe.*) is a different product; here we validate software and method.

VI. RESULTS AND VALIDATION

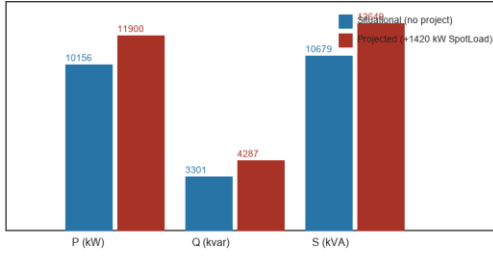
Diagnostics: $N_{msg}=0$. Precision: 11/11 OK, $p\text{-hat}=1.00$ (Clopper-Pearson 95% CI [0.72, 1.00]), $MAE=RMSE=0$. Allocation: $eps_P \approx 7.6 \times 10^{-5}\%$. SpotLoad 1420 kW; phase imbalance 0. Dual LF: $\Delta P=+1744$ kW ($\delta P=+17.17\%$), $\Delta Q=+986$ kvar, $\Delta S=+1969.7$ kVA; $V_{head}=99.84\%$ both cases. Closure 21/21.

TABLE I. CUSTOMER PRECISION (SAMPLE)

Customer	SED	Pot kW	Status
Caliza cementos	SE30940	3800.57	OK
Minerales Paracas	SE30480	1273.76	OK
LJMetales	SE30642	956.49	OK

Emsalsa	SE30353	458.00	OK
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Fig. 6. PA217 load-flow demand: situational vs projected



Source: PA217 COM LoadFlow narrative (Electro Dunas / RECYM)

Fig. 3. Situational vs projected demand (PA217).



Fig. 4. SpotLoad 1420 kW location (campaign).

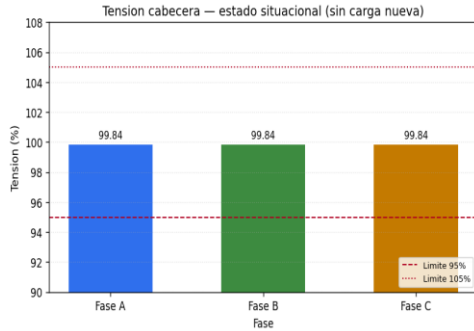


Fig. 5. Situational voltage map.

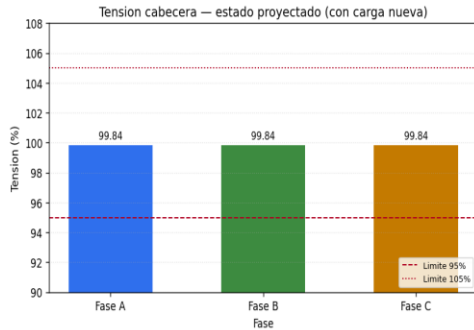


Fig. 6. Projected voltage map.

RECYM formalizes architecture, allocation/dual-LF models, pseudocode, and validation on a 97-feeder catalog, with PA217 as sample. Future work: fleet KPIs, official IEEEtran camera-ready, and AMI.

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VII. CONCLUSION